# Hardware CWE™ Special Interest Group (SIG)

**Chair:** Bob Heinemann (MITRE)

**Co-Chair:** “Manna” Parbati Kumar Manna (Intel)

MITRE Team: Steve Christey Coley, Alec Summers

**MITRE**

**January 16, 2026**

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# Agenda

**REMINDER: This meeting is being recorded.**

| | | | |
|---|---|---|---|
| 1 | RTL WG Update | Bob | 5 min |
| 2 | HW CWE Completeness | Steve | 30 min |
| 3 | 2026 Goals for SIG | Bob | 15 min |



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# Housekeeping

- **Schedule:**
  - **Next Meeting February 20:**
  - **Meeting is now the third Friday each month**
    - **12:30 – 1:30 PM EST (16:30 – 17:30 UTC)**
    - **Microsoft Teams**
- **Contact: cwe@mitre.org**
- **Mailing List:** *hw-cwe-special-interest-group-sig-list@mitre.org*
- **Minutes from previous meetings available on our GitHub site:**
  - **https://github.com/CWE-CAPEC/hw-cwe-sig**



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# Announcements

- **Out-of-Band CWE 4.19.1 release planned for January 20**
  - Only related to the CWE Top 25
- **CWE Content Development Repository (CDR) is now fully public.**
  - Enables the broader community to view, track, and contribute to entries submissions.
  - Content suggestions begin with the CWE Submission Form:
    - https://cwe.mitre.org/community/submissions/overview.html
  - CDR can be accessed here:
    - https://github.com/CWE-CAPEC/CWE-Content-Development-Repository/



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# RTL WG Updates

- **1st meeting December 8**
- **2nd meeting scheduled for January 26**
- **2 co-chairs; Vinod Viswanath and Jagadish Nayak**
- **Discussed and agreed on scope**
  - Scope focuses on RTL design and coding issues that have a security impact
    - Examples: FSMs, counters, FIFOs, registers, auto-generated RTL
  - Simulation–synthesis mismatches that introduce or expose security problems
  - Differences between simulated behavior and physical behavior that weaken or break security properties
  - Both logical and physical attack scenarios are considered relevant
  - CWE remains threat-neutral (does not assume a specific attacker model or threat type)



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# HW CWE Completeness



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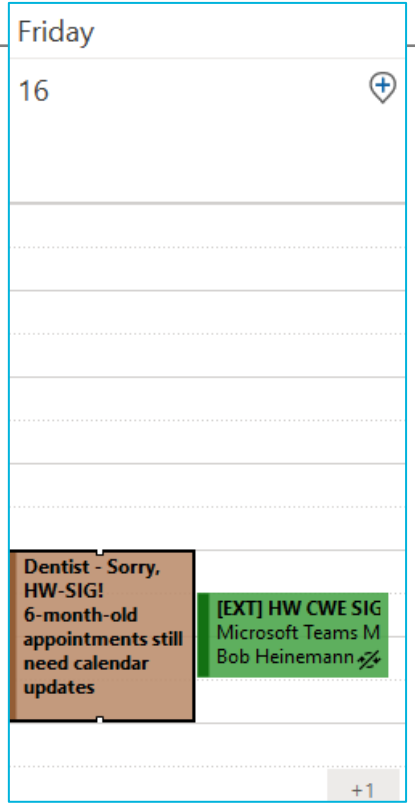
# An Apology from Steve.

- Root Causes: Missing Addition of High-Priority Event to Calendar (CWE-1250), combined with Failure to Recognize Appointment Reminder as Error Condition (CWE-754)
- Technical Impacts:
  - Financial (cancellation fees)
  - DoS (Denial of Speaking to the SIG for 20 minutes straight)
- I accept the reputational risk, but rest assured that I am paying the price in pain
- Apologies to Bob especially



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# What is “Completeness?”

- **Conceptual Coverage: CWE corpus has entries for all weakness-related concepts**
  - Minimize gaps (at least to the Base level?)
  - CWEs for weaknesses within scope of the CWE program (e.g., hardware, AI)
- **Individual CWE entries**
  - Entry has all “high-priority” top-level fields (elements) filled
  - Entry has all “preferred” elements filled, where possible
  - Entry has other elements filled as-needed
- **Breadth: all entries have the element**
- **Depth: all elements have full details**
- **Correctness / Quality**



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# Aspects of Individual-Entry Completeness

- **Can be evaluated relative to a set of CWEs (all; MIHW; HW view; Top 25; etc.)**
- **Breadth: % of entries with the top-level element**
  - For some elements, we want to MINIMIZE their presence (maintenance notes, extended description)
- **Depth: % of sub-elements filled (where feasible)**
  - Some elements are complex with multiple sub-elements
  - All sub-elements are filled (where feasible)
  - Elements that can have multiple instances (e.g., mitigations, consequences) “should” have all possible members (where feasible)
- **Correctness/Quality**
  - Data is correct (for some definition of “correct”) without any key omissions
  - Element meets “quality” expectations
- **Expectations might vary relative to abstraction (Pillar, etc.) and/or use case**



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# “Depth” Examples

- **We don’t have metrics for “depth” yet**
- **NOTE: numbers based on internal snapshot from November 4, 2025**
  - Many changes were made for CWE 4.19 (December 11, 2025)
- **Applicable Platforms: 738 weaknesses of 944**
  - 197 have a specific affected Language
    - 573 have a language “class” (not-language-specific, compiled, interpreted, assembly)
  - 231 with an affected Technology
- **Demonstrative Examples: 580 weaknesses of 944**
  - 327 CWEs only have 1 demox
  - 1323 “Example_Code” segments across all demox; 14 without Language specified
  - 238 CWEs with Java demox; 194 C; 30 Python; 4 Go; …
  - Community involvement could help here (e.g., Python secure coding project)
- **References – authors identified, URL is not 404, etc.**
- **Most Potential Mitigations don’t have “Effectiveness”**



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# Element Priority Varies by Task / User Type

- **Mapping? Don't care about code examples**
- **Many elements are "only" useful to developers and/or product security managers**
- **The UEWG spent time discussing weakness element importance as part of creating the current content filters (2023)**
- **Old worksheet (early 2010's) ranked elements based on stakeholder "tiers" and importance to those stakeholders:**
  - https://cwe.mitre.org/data/reports/stakeholder_field_priorities.html



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# Current Goals – Completeness Required (Subject to Change and Your Feedback)

| Element | Completeness Goals | Mapper | Developer | Bug Hunt | Academi | PSIRT |
|---|---|---|---|---|---|---|
| Name | Yes | High | High | High | High | High |
| Description | Yes | High | High | High | High | High |
| Diagram | Yes | - | Medium | Low | Low? | Medium |
| Related Weaknesses | Yes | High | High | High | High | High |
| Applicable Platforms | Yes | - | High | Low | Low | High |
| Modes of Introduction | Yes | - | High | Low | Medium | High |
| Common Conseq | Yes | - | High | Medium | Low | High |
| Detection Methods | Yes | - | High | Medium | Low | High |
| Potential Mitigations | Yes | - | High | Low | - | High |
| Demonstrative Examples | Yes | - | High | High | ? | Medium |
| Observed Examples | Yes - when available | - | High | High | Medium | Medium |
| References | Yes | - | Medium | High | High | Medium |
| Mapping Notes | Yes | High | - | High | Low | Medium |
| Content History | Yes | - | - | - | - | - |

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# Current Goals – Completeness Not Required (Subject to Change)

| Element | Completeness Goals | Mapper | Developer | Bug Hunt | Academi | PSIRT |
|---|---|---|---|---|---|---|
| Extended Description | No - avoid if possible | Low | Medium | Low | Medium | Medium |
| Weakness Ordinalities | No | Medium | Medium | Low | Low | Medium |
| Background Details | No | - | Low | - | - | Low |
| Alternate Terms | No - when available | High | Medium | High | Low | High |
| Functional Areas | No | Low | Medium | Low | Low | Medium |
| Affected Resources | No | Low | Medium | Low | Low | Low |
| Taxonomy Maps | No | Medium | - | - | Medium | Medium |
| Notes - Terminology | No - when available | High | Medium | Medium | Low? | Medium |
| Notes - Research Gaps | No | - | - | - | Medium | - |
| Notes - Applicable Platform | No - when available | - | Medium | Low | - | Medium |
| Notes - Maintenance | No - avoid if possible | - | - | - | - | - |
| Notes - Theoretical | No | Low | - | - | Low | - |
| Notes - Other | No - avoid if possible | - | - | - | - | - |
| Exploitation Factors | No - Discontinued | - | - | - | - | - |
| Likelihood Of Exploit | No - Discontinued | - | High | Medium | - | High |
| Attack Patterns | No - Discontinued | - | Medium | Low | - | Medium |

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# Completeness Status – as of CWE 4.19 (disclaimer: not all elements are listed)

| | All Active Weaknesses | | | View-1194 | | | MIHW | | |
|---|---|---|---|---|---|---|---|---|---|
| Field | # Filled | # Unfilled | % Filled | # Filled | # Unfilled | % Filled | # Filled | # Unfilled | % Filled |
| **Name** | 944 | 0 | 100% | 110 | 0 | 100% | 16 | 0 | 100% |
| **Description** | 944 | 0 | 100% | 110 | 0 | 100% | 16 | 0 | 100% |
| Extended_Description | 585 | 359 | 62% | 104 | 6 | 94% | 15 | 1 | 93% |
| **Diagram** | 75 | 869 | 8% | 2 | 108 | 1% | 0 | 16 | 0% |
| **Relationships** | 944 | 0 | 100% | 110 | 0 | 100% | 16 | 0 | 100% |
| **Common_Consequences** | 944 | 0 | 100% | 110 | 0 | 100% | 16 | 0 | 100% |
| **Applicable_Platforms** | 944 | 0 | 100% | 110 | 0 | 100% | 16 | 0 | 100% |
| **Modes_of_Introduction** | 944 | 0 | 100% | 110 | 0 | 100% | 16 | 0 | 100% |
| **Detection_Methods** | 493 | 451 | 52% | 44 | 66 | 40% | 15 | 1 | 93% |
| **Potential_Mitigations** | 670 | 274 | 71% | 106 | 4 | 96% | 16 | 0 | 100% |
| **Demonstrative_Examples** | 592 | 352 | 63% | 96 | 14 | 87% | 16 | 0 | 100% |
| **Observed_Examples** | 549 | 395 | 58% | 51 | 59 | 46% | 14 | 2 | 87% |
| Weakness_Ordinalities | 937 | 7 | 99% | 110 | 0 | 100% | 16 | 0 | 100% |
| Likelihood_of_Exploit | 183 | 761 | 19% | 2 | 108 | 2% | 0 | 16 | 0% |
| **References** | 581 | 363 | 62% | 78 | 32 | 70% | 16 | 0 | 100% |
| Mapping_Notes | 944 | 0 | 100% | 110 | 0 | 100% | 16 | 0 | 100% |
| Terminology_Notes | 35 | 909 | 4% | 2 | 108 | 1% | 0 | 16 | 0% |
| Alternate_Terms | 102 | 842 | 11% | 2 | 108 | 1% | 0 | 16 | 0% |
| Related_Attack_Patterns | 330 | 614 | 35% | 91 | 19 | 82% | 13 | 3 | 81% |
| Relationship_Notes | 132 | 812 | 14% | 6 | 104 | 5% | 3 | 13 | 18% |
| Taxonomy_Mappings | 633 | 311 | 67% | 11 | 99 | 10% | 1 | 15 | 6% |

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# Prioritizing Completeness Improvements

- **Q1: Should we choose a specific element, or a set of elements?**
- **Q2: Should we prioritize a specific task or use case over others?**
- **Q3: Which CWE(s) do we focus on?**
  - MIHW, all hardware view, whatever people are interested in…
- **Q4: Should we focus more on depth than breadth?**
  - Affected languages (and which ones?); more variety in demonstrative examples; effectiveness estimates for mitigations/detection; etc.
- **Q5: How can the HW-SIG help?**
  - Gauge interest and availability
  - Reports can be provided in the coming weeks
  - Consider CDR submissions to modify individual entries



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# HW CWE SIG: Help Set Our Focus

**For the next 3 minutes, please answer these in chat.**

**Start your answers with Q1 or Q2 so we can sort them later.**

**PART 1:**

- **Q1 – Outcomes (6–12 months)**
  - What is one concrete outcome you'd love this SIG to achieve?
- **Q2 – Themes**
  - What 1–2 HW CWE themes or problem areas are most important for us to tackle?

**PART 2:**

- **Q3 – Ownership**
  - Is there a theme/outcome you'd be willing to Champion or co-champion?
  - Contribute to in a small working group?



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# Next Meeting (February 20)

**CWE@MITRE.ORG**

- **Mailing List:** *hw-cwe-special-interest-group-sig-list@mitre.org*
  - ***NOTE: All mailing list items are archived publicly at:***
    - *https://www.mail-archive.com/hw-cwe-special-interest-group-sig-list@mitre.org/*
- **What would members of this body like to see for the next HW SIG agenda?**
- **Questions, Requests to present? Please let us know.**



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